



A Pangenome Sequence Search and Coordinate Translation Service for the UCSC Genome Browser

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Motivation

The HPRC v2.0 minigraph—Cactus graph holds **233 samples / 464 haplotypes** — but there has been no interactive way to query it.

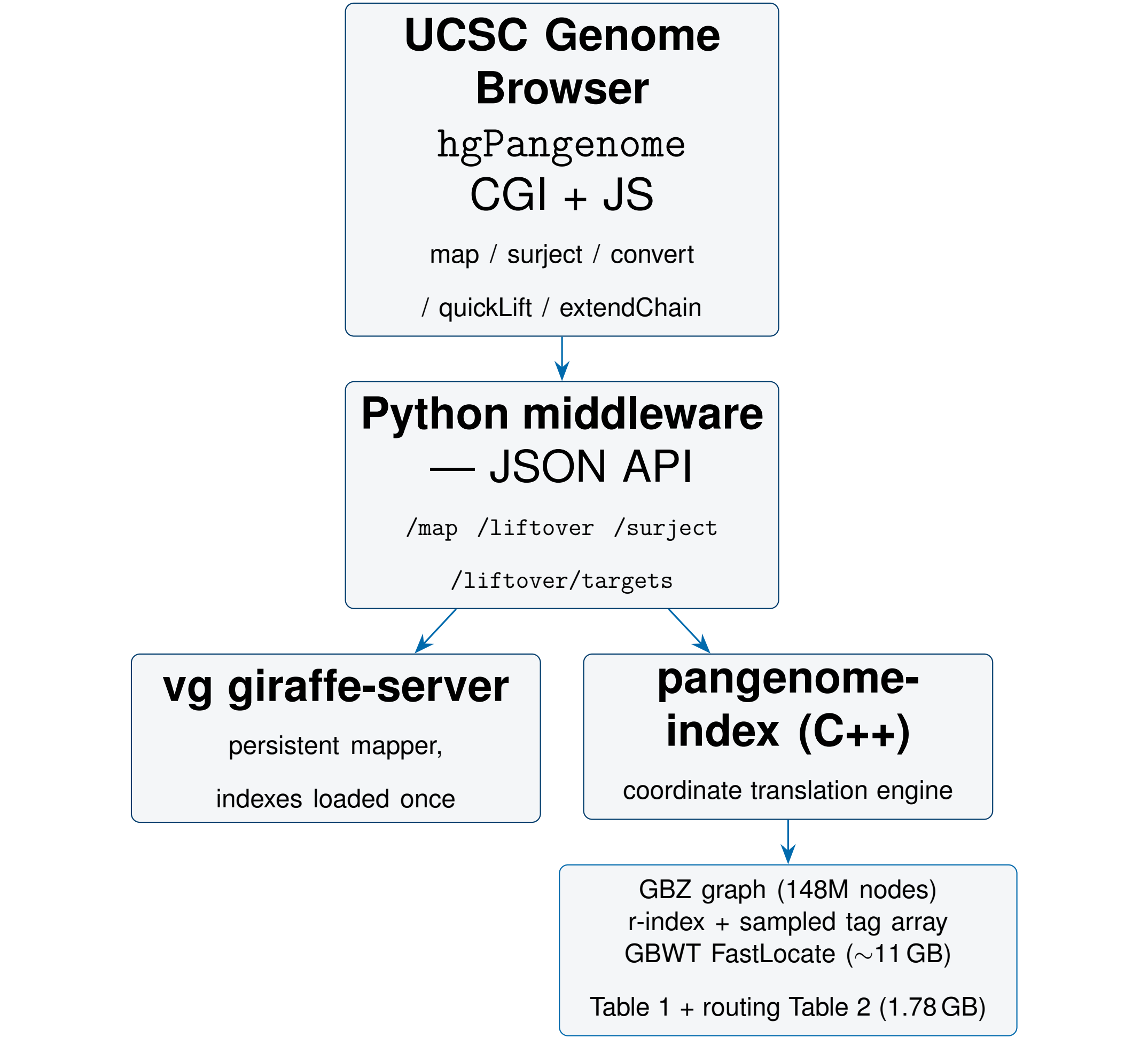
- ▶ A researcher with a **sequence** cannot ask which haplotypes carry it, or where it lies on each of them.
- ▶ A researcher with a **position** cannot convert it to another assembly: chain files cover only **56** of the **~213,000** possible sample-to-sample pairs.

We built a service that answers both through the graph itself, integrated into the UCSC Genome Browser.

Three capabilities

1. **Pangenome sequence search** — a BLAT-like page: a pasted sequence is mapped to the whole graph with `vg giraffe`. Results show every assembly carrying the exact allele path, all 464 haplotypes ranked by identity, and one-click positions on any haplotype, drawn as a browser track with base-level mismatch coloring.
2. **Any-to-any coordinate translation** — liftover between any two of the 466 assemblies at interactive speed, with a scored “which assemblies contain this region” destination picker.
3. **Annotation lifting** — a QuickLift chain generated *at request time*, so users land on an HPRC assembly with CHM13’s genes, RefSeq, CenSat and T2T Encode at translated positions, plus an **Alignment Differences** track marking indels and mismatches between the two assemblies.

Architecture



Core browser modifications: **3 files / 15 lines**; everything else is a new, self-contained CGI, dark unless configured.

Translation through the graph

1. **Table 1**: `contig:start-end` → graph path + path-local interval.
2. **Candidate discovery**: probe nodes on the source path (~1 / 2 kb); each GBWT probe returns *every* haplotype on that node. The **routing Table 2** replaces probing with a direct lookup.
3. **Anchoring**: nodes visited exactly once by both sides are guaranteed orthologous; a colinearity gate rejects implausible pairings.
4. **Trace**: an LF-walk between first and last shared node emits per-base correspondences.

Results fold into **chain-semantics blocks**: a block breaks only on indels, orientation flips, or contig changes — never on substitutions.

Routing table: 17.2 TB → 1.78 GB

An all-pairs table over the frequency-filtered graph (224M path fragments, 2,367× fragmentation) would cost **17.2 TB**. Building over the non-filtered graph (94,554 paths) and storing *routing only* — which targets a source range touches, never their coordinates — eliminates every failure mode of interval pairing:

	interval-pairing	routing
abandoned runs	5,025,344	0
splits	35,734,985	0
segments	68.2 M	45.7 M
size	3.69 GB	1.78 GB
build (32 thr)	—	35 min

Coverage and identity, all 464 at once

One pass over a mapped read’s nodes scores **every haplotype** simultaneously (each probe already returns all of them):

$$\text{coverage}(H) = \frac{\text{aligned bases on nodes } H \text{ visits}}{\text{aligned bases}}$$

$$\text{identity}(H) = \frac{\dots \text{that also } \textit{matched} \text{ there}}{\text{aligned bases}}$$

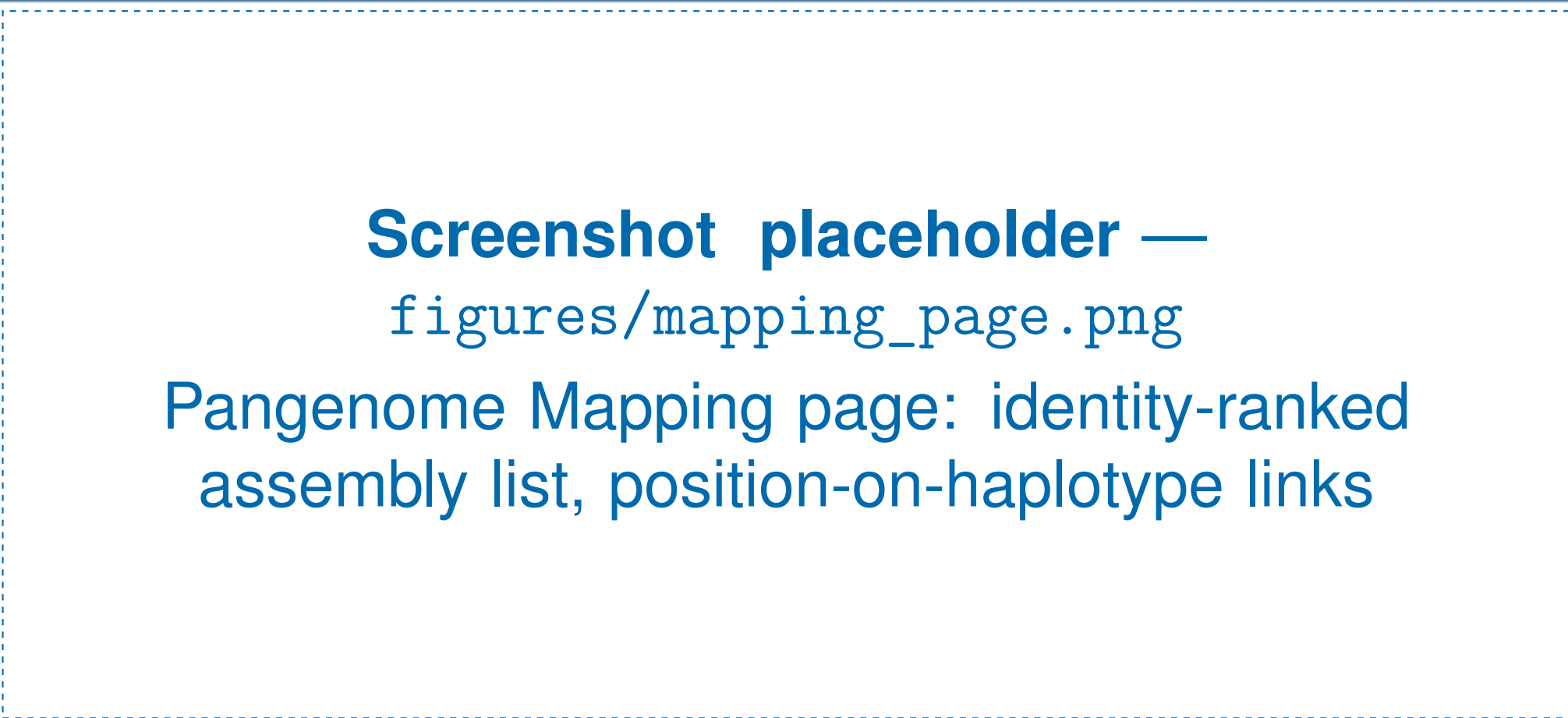
$\text{identity} \leq \text{coverage}$ always; the gap is the mismatch burden. Shown together, they separate “present but diverged” from “half missing” — which one score conflates.

On-the-fly QuickLift chains

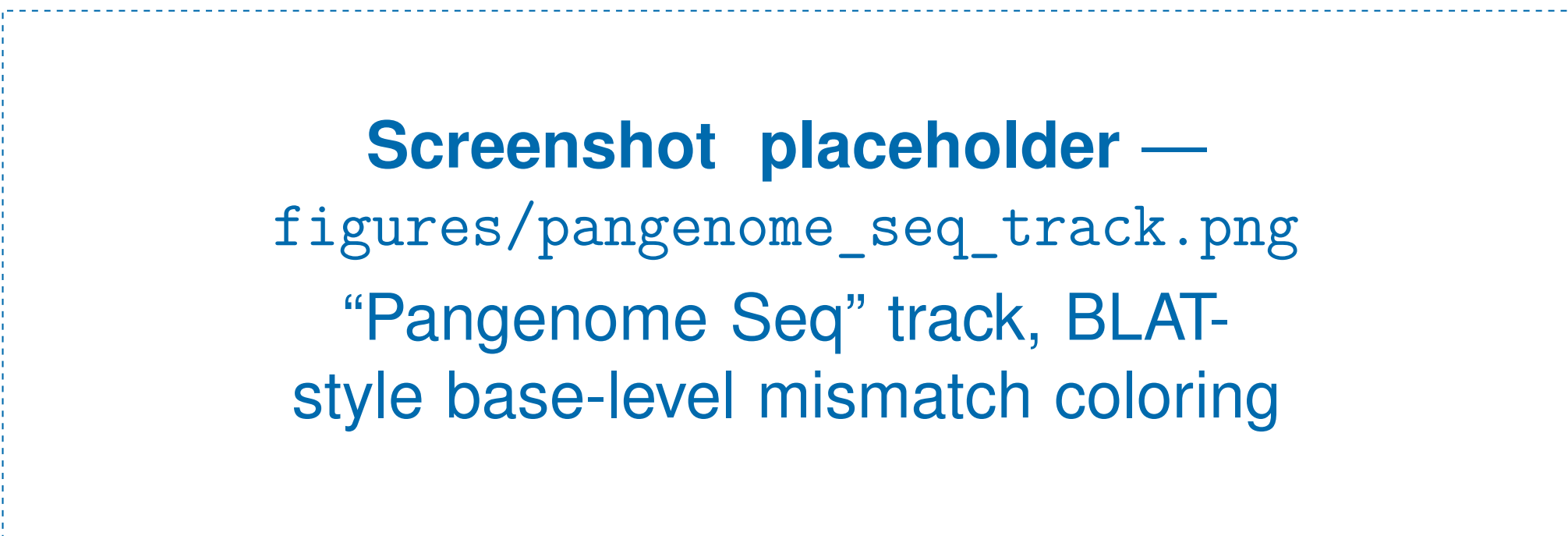
Blocks become `bigChain+bigLink` on the fly (JSON → BED → `bedToBigBed`), served through a generated hub — QuickLift’s renderer needs nothing else. **Two-phase padding** makes panning work:

- ▶ **Phase 1** (synchronous, 0.7 s): pad to 5× span, clamped to 250 kb–1 Mb.
- ▶ **Phase 2** (background, 11 s nobody waits for): a 222-byte `sendBeacon` during navigation rebuilds at ±5 Mb; the next pan picks it up, no reload.

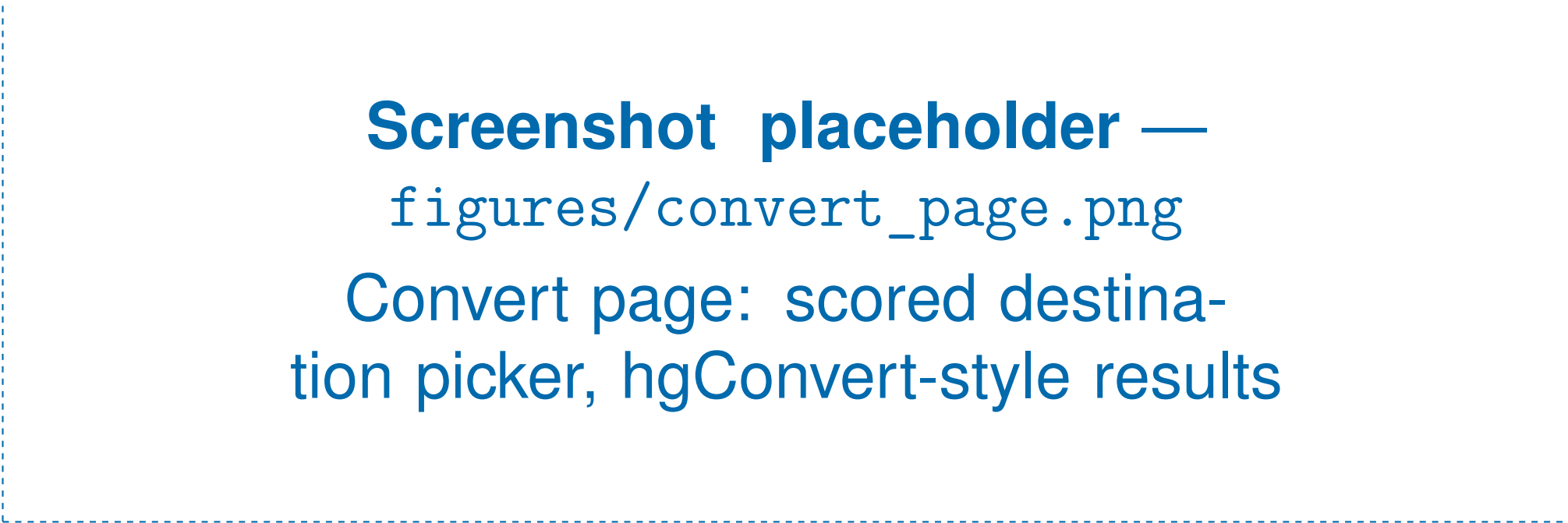
Sequence search in the browser



A 300 bp query scores all 464 haplotypes in a couple of seconds. Clicking a haplotype re-surjects the cached alignment — no remapping — and the browser opens there with the sequence drawn as a track:



Conversion and lifted annotations



Pangenome assemblies also appear in the stock `hgConvert` dropdown (462 destinations from a sample assembly). Converting lands on the target with **annotations lifted from CHM13** and Alignment Differences:



Verified base-by-base: across 22 kb of ABO, the track shows exactly the 2 real C→T mismatches between `hs1` and `HG00597#1`.

Chain widening in action

Lifted gene items visible after panning away from the converted window (1 = track label only):

pan distance	phase 1 (1.1 Mb)	phase 2 (10 Mb)
converted window	3	3
~400 kb	129	129
~2 Mb	1	5
~4 Mb	1	31

Performance

Translation latency (warm, CHM13 chr10 source):

span	median	completeness
100 bp	22 ms	99.0%
1 kb	21 ms	99.8%
10 kb	115 ms	99.6%
100 kb	599 ms	97.2%
1 Mb	4.3 s	93.1%

Fixing candidate discovery (stop-at-first-hit → exhaustive) took 1 Mb completeness from **2.9%** to **93.1%**, and made it 2× faster.

Concurrency (3.4 kb queries, GIL released in C++):

threads	q/s	speedup
1	6.3	1.0×
4	23.6	3.7×
8	60.3	9.6×
32	65.3	10.4×

Scored destination picker: 100 kb region in **1.28 s** vs 29.5 s exact (23×). Chain generation: linear, ~1.14 s/Mb.

At a glance

464	~20 ms
haplotypes scored per query	gene-scale liftover
1.78 GB	15
routing table, not 17.2 TB	lines changed in kent core

Future work

- ▶ **Bulk offline chain generator**: direct path intersection instead of r-index probing — est. ~300× faster (~1.6 h per target genome-wide vs 19 days via the API) — for precomputed whole-chromosome chains and full QuickLift parity.
- ▶ **Persistent chain cache**, so widened chains outlive the one-hour session and popular regions become free after the first visit.
- ▶ Segdup-dense wide intervals lose positions (~46% at 1q21.1); the bulk path removes the probe spacing that misses them.
- ▶ Public release: core-change sign-off, per-install API tokens, service-side concurrency deployment.

Acknowledgements & availability

Built on `vg`, GBWT/GBZ and the tag-array index (Sirén et al.), the HPRC v2.0 minigraph—Cactus graph, and the UCSC Genome Browser (kent). `vg giraffe-server` is under review as `vgteam/vg PR #4879`. We thank the UCSC Genome Browser team and the Human Pangenome Reference Consortium. Poster sources: github.com/parsaeskandar/hprc_2026_poster